313314

12425

3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions :

(1) All Questions are *compulsory.*

(2) Answer each next main Question on a new page.

(3) Illustrate your answers with neat sketches wherever necessary.

(4) Figures to the right indicate full marks.

(5) Assume suitable data, if necessary.

(6) Use of Non-programmable Electronic Pocket Calculator is permissible.

(7) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

Marks

1. Attempt any FIVE of the following : 10

(a) State the necessity of frames in automobiles.

(b) Define the terms Chassis in automobile.

(c) List different clutch lining materials.

(d) State the principle on which friction clutch works.

(e) State the functions of gear box.

(f) State types of automobile gear boxes.

(g) State the effects of incorrect tyre inflation.

2. Attempt any THREE of the following : 12

(a) State different loads acting on frame.

(b) Describe working of single plate clutch with tis application.

(c) Explain with neat sketch working of transfer case.

(d) "Automatic transmissions are better option than manual transmission." Justify statement with suitable reasons.

3. Attempt any THREE of the following : 12

(a) Classify vehicle layout with respect to

(i) Location of engine

(ii) Arrangement of engine

(iii) Application

(iv) No. of live axles

(b) Describe the working of overdrive mechanism in automobiles.

(c) Describe with neat sketch working of torque tube drive.

(d) State the functions of universal joint and slip joint.

4. Attempt any THREE of the following : 12

(a) Describe working of FERWD layout with suitable example.

(b) Describe with neat sketch working of constant mesh gear box.

(c) Give tyre designation with one example and interpret the meaning of terms involved in it.

(d) Describe construction details of hollow and solid propeller shaft.

(e) State requirements of automobile clutches.

5. Attempt any TWO of the following : 12

(a) State necessity of differential on a vehicle. Draw neat sketch of the differential unit and label all parts.

(b) "The motorcycle uses multiplate clutch." Justify your answer comparing single plate clutch with multiplate clutch.

(c) Describe construction and working of CVT with application.

6. Attempt any TWO of the following : 12

(a) Sketch arrangement of following types of rear axles and give one application of each :

(i) Semi-floating

(ii) Full floating

(b) Describe construction and working of front wheel drive shaft.

(c) Compare tubed tyre and tubeless tyre.